

Effective connectivity accross hemispheres in children with callosal dysgenesis

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Abstract

This project investigates whether effective connectivity (EC) during sensorimotor processing differs between typically developing children (TDC) and children with corpus callosum dysgenesis (CCD). Task-fMRI data were acquired during a reaction-time paradigm in which each event contained either an auditory stimulus, a visual stimulus, or a combined audio-visual stimulus. Auditory and visual stimuli were presented on either the left or right side; for combined audio-visual events, both modalities were presented on the same side. Because each task also specified the response hand, every event could be classified as crossed or uncrossed according to the relationship between stimulus side and response side. Dynamic Causal Modelling (DCM) was used to compare biologically motivated EC architectures between bilateral visual, auditory, and motor regions. The model space tested whether crossed sensorimotor transfer was best explained by direct, motor-relay, sensory-relay, or baseline null pathways, and whether uncrossed processing was restricted to same-hemisphere sensory-to-motor pathways or required additional flexible interhemispheric candidate connections. Random-effects Bayesian model selection showed that the winning architectures belonged to the flexible uncrossed family, indicating that even uncrossed trials were better explained when additional candidate pathways between hemispheres were allowed. Parametric Empirical Bayes (PEB) analysis further showed group differences in both intrinsic and task-modulated connectivity parameters. Many intrinsic effects were intra-hemispheric and stronger in CCD, consistent with compensatory strengthening of within-hemisphere pathways, while task-modulated effects also involved interhemispheric visual and auditory pathways. Overall, the results suggest selective reweighting of intra- and inter-hemispheric sensorimotor pathways in CCD rather than a complete loss of left-right communication.

tional MRI acquired in pediatric participants as part of the *Swiss Pre- to Postnatal Neurodevelopmental Cohort*. Participants responded as quickly as possible to auditory or visual stimuli by pressing a response button with the designated hand. The experiment included ipsilateral conditions, in which the sensory stimulus and motor response occurred on the same side, and contralateral conditions, in which the stimulus and response occurred on opposite sides. This design makes it possible to compare brain responses and connectivity according to stimulus modality, response side, and interhemispheric transfer demand.

The objective of this project is to test whether EC during sensorimotor processing differs between TDC and CCD children (see figure 1). The corpus callosum is the main commissural pathway connecting the two cerebral hemispheres and plays a central role in bilateral integration, interhemispheric communication, and coordination of sensory and motor information. In CCD, this pathway is partially or completely altered. However, previous clinical and neuroimaging evidence suggests that functional interhemispheric communication may still persist through compensatory or alternative pathways [2], [3]. The crossed and uncrossed task conditions are therefore well suited for DCM, because they allow us to test whether interhemispheric communication is reduced, reorganized, or maintained in CCD compared with TDC.

The analysis focuses on bilateral primary motor cortex (M1), primary auditory cortex (A1), and primary visual cortex (V1), which capture the sensory input and motor output components of the task. The four task labels encoded the instructed response hand and the nominal stimulation block. MD refers to a right-hand response and MG to a left-hand response. During each task, the modality of each event was recorded — whether it was auditory-only, visual-only, or audio-visual — along with the side of stimulus presentation (left or right). In audio-visual events, the auditory and visual stimuli were always presented on the same side. Crossed and uncrossed conditions were subsequently determined at the event level by comparing the side of stimulus presentation with the instructed response hand.

1 Introduction

The dataset used in this study was obtained from the Translational Machine Learning Laboratory and is presented in [1]. It includes high-resolution structural MRI and task-based func-

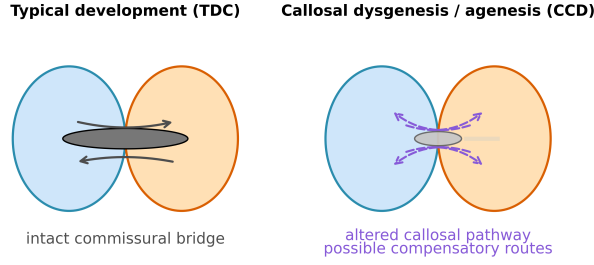


Figure 1: Schematic illustration of interhemispheric communication in typical development and callosal dysgenesis/agenesis.

2 Methods

Functional MRI data were acquired with the following parameters: repetition time (TR) = 0.8 s, echo time (TE) = 30 ms, and spatial resolution = $2.4 \times 2.4 \times 2.4 \text{ mm}^3$. During acquisition, participants performed a reaction-time paradigm derived from the classical Poffenberger task [4]. The final analyzed cohort included 26 pediatric participants: 8 children with corpus callosum dysgenesis (CCD; 4 females, 4 males; age: 9.04 ± 1.48 years, range: 7.01–11.13 years) and 18 typically developing children (TDC; 11 females, 7 males; age: 9.97 ± 1.84 years, range: 6.73–12.79 years). Overall, the analyzed cohort included 15 females and 11 males, with a mean age of 9.68 ± 1.76 years.

2.1 Single-subject DCM

DCM was used to test hypotheses about directed EC between sensory and motor regions during the task. DCM is a hypothesis-driven generative modelling framework: instead of estimating undirected statistical associations, it evaluates models of how experimental inputs drive hidden neuronal activity and how this activity gives rise to the measured BOLD signal [5, 6]. The hidden neuronal state vector $x(t)$ evolves according to intrinsic connectivity, experimental inputs, and task-dependent modulations:

$$\dot{x}(t) = \left(A + \sum_j u_j(t) B^{(j)} \right) x(t) + C u(t).$$

Here, A represents baseline EC, $B^{(j)}$ represents the modulation of connectivity by the j -th experimental condition, and C specifies where external inputs enter the network. Predicted neuronal activity is then passed through a hemodynamic observation model,

$$y(t) = h(x(t), \theta_h) + \epsilon,$$

to generate predicted BOLD responses. Competing architectures were compared using model evidence, approximated by the variational free energy F , which balances accuracy and complexity.

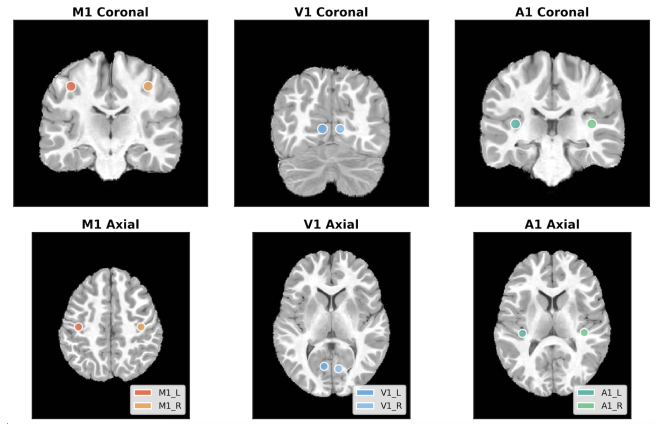


Figure 2: Spherical ROI masks overlaid on the preprocessed T1 image of sub-03. Functional MRI data were acquired with the following parameters: repetition time (TR) = 0.8 s, echo time (TE) = 30 ms, spatial resolution = $2.4 \times 2.4 \times 2.4 \text{ mm}^3$

Mean BOLD time series were extracted from six spherical ROI masks (bilateral M1, A1, and V1) defined as 5-mm-radius spheres centered on peak coordinates in MNI space, derived from NeuroVault activation maps [7]. For each ROI, the binary mask was resampled to the subject’s BOLD grid when necessary, voxels with mask value greater than zero were selected, and the mean BOLD signal was extracted and z-scored before DCM estimation. (See figure 2)

Table 1: ROI centers used for DCM time-series extraction. Coordinates are in MNI space in millimetres.

ROI	x	y	z	Radius (mm)
M1_L	-40	-20	49	5
M1_R	38	-20	49	5
A1_L	-37	-28	11	5
A1_R	40	-27	11	5
V1_L	-9	-69	6	5
V1_R	9	-72	6	5

The intrinsic matrix A encoded self-connections and same-hemisphere sensory-to-motor connections: V1_L and A1_L to M1_L, and V1_R and A1_R to M1_R. The driving input matrix C specified that visual inputs entered V1 and auditory inputs entered A1 according to stimulation side. The modulatory matrices B encoded task-dependent changes in connectivity associated with crossed and uncrossed conditions.

An eight-model space was constructed to compare alternative hypotheses about crossed sensorimotor transfer and uncrossed processing. Four crossed hypotheses were tested: NULL, RELAY, DIRECT, and SENSORY_RELAY. The NULL model did not mean absence of activity or connectivity; it was a control architecture in which no additional crossed interhemispheric route was added beyond local sensory-to-motor modulation. The RELAY model assumed transfer through local motor cortex before contralateral motor cortex. The DIRECT model assumed a direct sensory-to-contralateral-motor pathway. The SENSORY_RELAY model assumed

transfer through contralateral sensory cortex before reaching contralateral motor cortex (Figure 3).

Each crossed hypothesis was combined with two uncrossed connectivity constraints: `INTRA_ONLY` and `FLEX`. In `INTRA_ONLY`, uncrossed conditions were constrained to same-hemisphere sensory-to-motor pathways. In `FLEX`, additional candidate pathways were allowed, including sensory-to-sensory, sensory-to-contralateral motor, motor-to-motor, and contralateral sensory-to-motor connections. This resulted in 4 crossed hypotheses $\times$ 2 uncrossed configurations = 8 DCM architectures (see figure 3). A fully saturated model containing all possible pathways was not included because the goal was to compare biologically interpretable hypotheses rather than estimate an unconstrained graph. This is consistent with the hypothesis-driven use of DCM, where Bayesian model evidence penalizes unnecessary complexity [8–10].

For each subject and task, all eight DCMs were estimated using SPM12 following the standard first-level DCM framework for fMRI [6]. At the single-subject level, posterior model probabilities were computed from the free energy values using a softmax transformation over the eight models.

2.2 Group-level model comparison: RFX BMS

After single-subject DCM estimation, random-effects Bayesian model selection (RFX BMS) was used to compare architectures across subjects. The group-level script reads the subject- and task-specific free energy values saved by SPM12 in the `F_values.mat` files. Since F approximates log model evidence, it can be used to compare models while accounting for both fit and complexity [11, 12].

For each subject and task, posterior model probabilities were computed as:

$$p(m_i|y) = \frac{\exp(F_i)}{\sum_{k=1}^K \exp(F_k)},$$

where $K = 8$ is the number of models. At the group level, RFX BMS estimates the expected model frequency and exceedance probability, allowing different subjects to be best explained by different models. This is more appropriate than fixed-effects comparison for this project, because CCD and TDC children may rely on different or more variable pathways [11, 13].

Family-level inference was also performed. Models were grouped by crossed pathway hypothesis (`NULL`, `RELAY`, `DIRECT`, `SENSORY_RELAY`) and by uncrossed configuration (`INTRA_ONLY`, `FLEX`). Family evidence was computed using a log-sum-exp operation:

$$F_{\mathcal{F}} = \log \sum_{m_i \in \mathcal{F}} \exp(F_i),$$

allowing broader hypotheses to be tested, such as whether flexible uncrossed models are preferred over strictly intra-hemispheric models [11, 13].

2.3 Parametric Empirical Bayes

After RFX BMS, PEB was used to test whether specific EC parameters differed between CCD and TDC children. Following the hierarchical framework of Zeidman et al. [12], RFX BMS first identified the best-supported architecture, and PEB then tested group effects on the estimated DCM parameters. The analysis focused on the dominant flexible architecture, `SENSORY_RELAY__UNC_FLEX`.

For each task, 38 parameters were tested for the `CCD_minus_TDC` contrast: 10 intrinsic A parameters and 28 task-modulated B parameters. Across four tasks, this yielded 152 tested group-effect parameters. The number of parameters follows from the switched-on DCM matrices: the intrinsic A matrix contained six self-connections and four same-hemisphere sensory-to-motor connections, while the B matrices contained four uncrossed conditions with five switched-on connections each and four crossed conditions with two switched-on connections each (Appendix C).

For each task, estimated DCMs from all available subjects were entered into a second-level design matrix:

$$X = \begin{bmatrix} 1 & g_1 \\ \vdots & \vdots \\ 1 & g_N \end{bmatrix}, \quad g_i = \begin{cases} -1, & \text{TDC} \\ +1, & \text{CCD}. \end{cases}$$

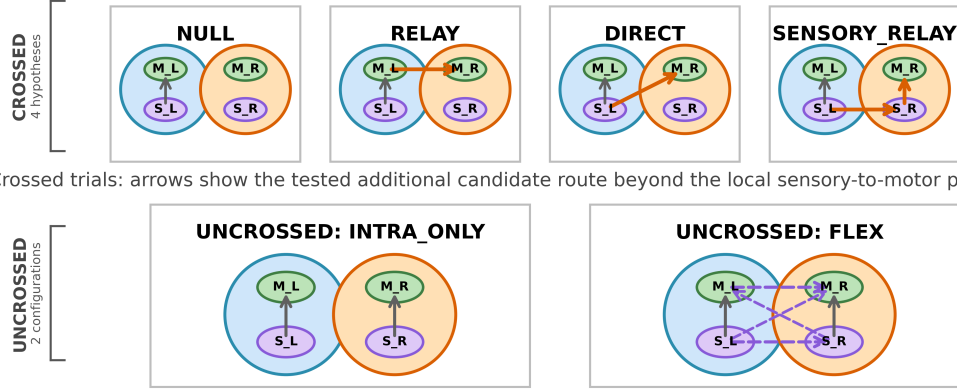
The first column estimates the group mean, while the second estimates the `CCD_minus_TDC` contrast. A positive PEB estimate therefore indicates a stronger parameter in CCD, whereas a negative estimate indicates a stronger parameter in TDC. PEB was applied to both intrinsic A parameters and modulatory B parameters. Bayesian model reduction and Bayesian model averaging were then used to identify parameters with strong evidence for a group effect, with posterior probabilities $Pp \geq 0.95$ interpreted as strong evidence.

3 Results

3.1 Single-subject DCM and RFX BMS results

Single-subject DCM estimation was completed for the full cohort of 26 participants across the four tasks. Across subjects and tasks, winning models consistently belonged to the `UNC_FLEX` family (see figure 4), meaning that even uncrossed trials were better explained when additional candidate interhemispheric pathways were allowed. This suggests that sensorimotor processing in this task is not purely local and that bilateral interactions remain relevant even when the stimulus-response mapping does not explicitly require inter-hemispheric transfer.

RFX BMS confirmed this pattern at the group level. The `FLEX` family received very strong evidence compared with the `INTRA_ONLY` family, with an expected frequency of approximately 0.975 and an exceedance probability of 1.0. This indicates that flexible uncrossed architectures were strongly



Crossed trials: arrows show the tested additional candidate route beyond the local sensory-to-motor pathway.

Model space = 4 crossed hypotheses x 2 uncrossed configurations = 8 DCM architectures

Figure 3: Eight-model DCM used for subject-level model comparison. The upper row represents the four hypotheses tested for crossed sensorimotor transfer: NULL, RELAY, DIRECT, and SENSORY_RELAY. Orange arrows indicate the additional crossed candidate pathway introduced by each model beyond the local sensory-to-motor pathway. The lower row represents the two configurations tested for uncrossed trials: INTRA_ONLY, where processing is constrained to same-hemisphere sensory-to-motor pathways, and FLEX, where additional inter-hemispheric candidate pathways are allowed. Combining the four crossed hypotheses with the two uncrossed configurations gives eight candidate DCM architectures per subject and task.

preferred over strictly same-hemisphere sensory-to-motor models.

For crossed-transfer hypotheses, relay-type architectures were preferred. The dominant model was SENSORY_RELAY__UNC_FLEX, with an expected frequency of approximately 0.542 and an exceedance probability of 0.987. The RELAY__UNC_FLEX architecture was the second most frequent model, while direct crossed transfer received comparatively little support (Figure 4).

by models allowing additional interhemispheric candidate pathways during uncrossed processing. However, the crossed-transfer component differed between groups. In CCD children, the dominant winners were evenly split between SENSORY_RELAY__UNC_FLEX and RELAY__UNC_FLEX (4/8 subjects for each architecture). In contrast, TDC children showed a clearer preference for SENSORY_RELAY__UNC_FLEX (14/18 subjects), with fewer subjects best explained by RELAY__UNC_FLEX (4/18 subjects). This shows that TDC children rely more consistently on a sensory-relay architecture, in which crossed transfer is mediated through contralateral sensory regions before reaching motor cortex. CCD children, by contrast, show a more heterogeneous pattern, with some subjects following a sensory-relay organization and others favoring a motor-relay pathway.

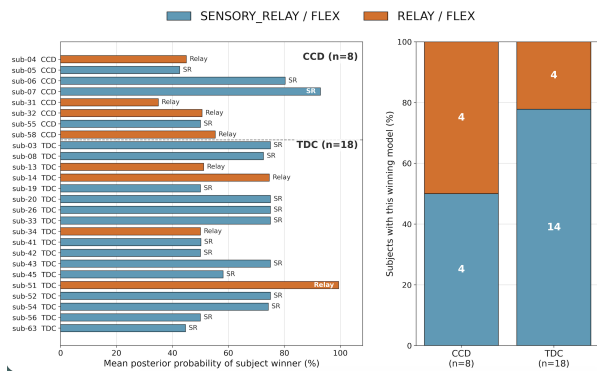


Figure 4: Subject-level dominant winning DCM architectures in the full cohort. Bars show the mean posterior probability of each subject's winning architecture across tasks, and the stacked bars summarize the proportion of CCD and TDC subjects assigned to each dominant winner.

The subject-level summary in Fig. 4 shows that all dominant winning architectures belonged to the UNC_FLEX family, meaning that both groups were better explained

3.2 PEB results

The PEB analysis tested whether EC parameters differed between CCD and TDC children within the dominant architecture, SENSORY_RELAY__UNC_FLEX. Across the four tasks, 27 out of 152 tested CCD_minus_TDC parameters showed strong evidence group effects ($P_p \geq 0.95$). These included 16 intrinsic *A* effects and 11 task-modulated *B* effects (Figure 8).

The intrinsic *A* effects were exclusively intra-hemispheric. Thirteen were more positive in CCD, compared with three more positive in TDC (see figure 5). This supports the hypothesis that CCD is associated with altered within-hemisphere baseline dynamics. [14] However, many of these

effects involved self-connections, which in DCM reflect self-inhibition rather than simple excitatory coupling [6]. Therefore, positive CCD effects on self-connections should be interpreted as increased self-inhibition in CCD, not simply as stronger excitatory connectivity.

Task-modulated B effects showed a more mixed pattern. Among the 11 strong evidence B effects, five were more positive in CCD and six were more positive in TDC. Intra-hemispheric modulations were present in both groups, with five effects more positive in CCD and three more positive in TDC. In contrast, the three evident interhemispheric modulations were all more positive in TDC (see figure 7).

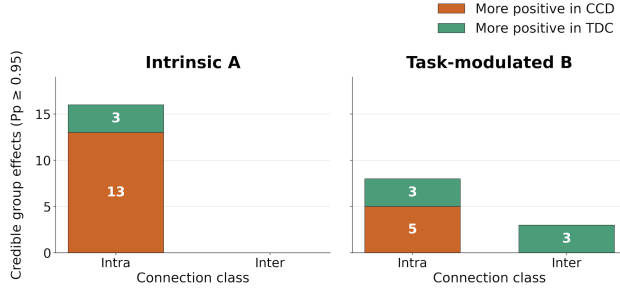


Figure 5: Effects with high posterior probability classified as intra- or inter-hemispheric for the full cohort (26 participants). Intrinsic A effects were exclusively intra-hemispheric and mostly more positive in CCD. Task-modulated B effects showed a mixed pattern, with interhemispheric modulations more positive in TDC. Self-connections are classified as intra-hemispheric; their sign reflects self-inhibition.

4 Discussion

The full-cohort results support the hypothesis that CCD is associated with a reorganization of sensorimotor EC, rather than a complete disruption of functional communication between hemispheres. At the architecture level, RFX BMS showed that the dominant winning models belonged to the UNC_FLEX family. This indicates that, even for uncrossed trials, the data were better explained when additional candidate interhemispheric pathways were allowed, rather than when processing was restricted to same-hemisphere sensory-to-motor pathways only. Therefore, both TDC and CCD participants appear to rely on flexible bilateral architectures during sensorimotor processing. However, the distribution of crossed-transfer models differed between groups: TDC participants more often favored SENSORY_RELAY architectures, whereas CCD participants showed a more balanced distribution between SENSORY_RELAY and RELAY models (Fig. 4). This suggests that crossed sensorimotor transfer is better explained by intermediate sensory or motor relay pathways than by a direct sensory-to-contralateral-motor route, and that CCD may be associated with more heterogeneous relay strategies. [3]

The PEB analysis refined this interpretation by testing group differences at the parameter level. The strongest group-related pattern was observed in the intrinsic A parameters. All

credible intrinsic group effects were intra-hemispheric, and most were more positive in CCD. Since the intrinsic A matrix in this model included self-connections and same-hemisphere sensory-to-motor connections, this result suggests that baseline within-hemisphere dynamics are altered in CCD. This is compatible with the hypothesis of intra-hemispheric compensation in callosal dysgenesis, and is consistent with Shi et al. [14], who reported increased intra-hemispheric structural connectome metrics in children with CCD. In the present task-fMRI analysis, a related compensatory pattern is observed at the level of EC, suggesting that altered callosal anatomy may be associated with stronger or differently regulated local sensorimotor coupling.

At the same time, the results do not support a purely intra-hemispheric compensatory mechanism. First, the strong preference for UNC_FLEX architectures indicates that interhemispheric candidate pathways were still needed to explain the data [15]. Second, the task-modulated B parameters showed credible group effects involving both intra- and inter-hemispheric connections. In contrast to the intrinsic effects, interhemispheric task-dependent modulations were more prominent in TDC. This suggests that TDC participants may rely more consistently on flexible task-dependent communication between hemispheres, whereas CCD participants show a stronger contribution of baseline intra-hemispheric reorganization and a more variable pattern of relay strategies across tasks. [3].

Overall, these findings suggest that children with CCD do not simply replace interhemispheric communication with intra-hemispheric processing. Instead, CCD appears to involve a selective reweighting of sensorimotor pathways. At baseline, within-hemisphere EC is more strongly altered in CCD, consistent with possible compensatory strengthening or altered regulation of local pathways [3], [2], [15]. During task performance, however, group differences also involve bilateral pathways (Fig. 7). The main interpretation is therefore that CCD-related compensation is not purely local, but reflects a different balance between intra- and inter-hemispheric sensorimotor communication.

5 Conclusion

This project used DCM, RFX BMS, and PEB following papers [6] and [12] to investigate sensorimotor EC in children with CCD and TDC. The full-cohort results show that flexible architectures allowing additional interhemispheric candidate pathways best explained the data. At the parameter level, PEB revealed group differences with high posterior evidence in both intrinsic and task-modulated EC. Intrinsic effects were mainly consistent with intra-hemispheric reorganization in CCD, while task-modulated interhemispheric effects were more prominent in TDC. These findings support the interpretation that CCD is associated with compensatory reweighting of sensorimotor pathways rather than a complete loss of left–right functional communication [3], [2].

A Additional RFX BMS results

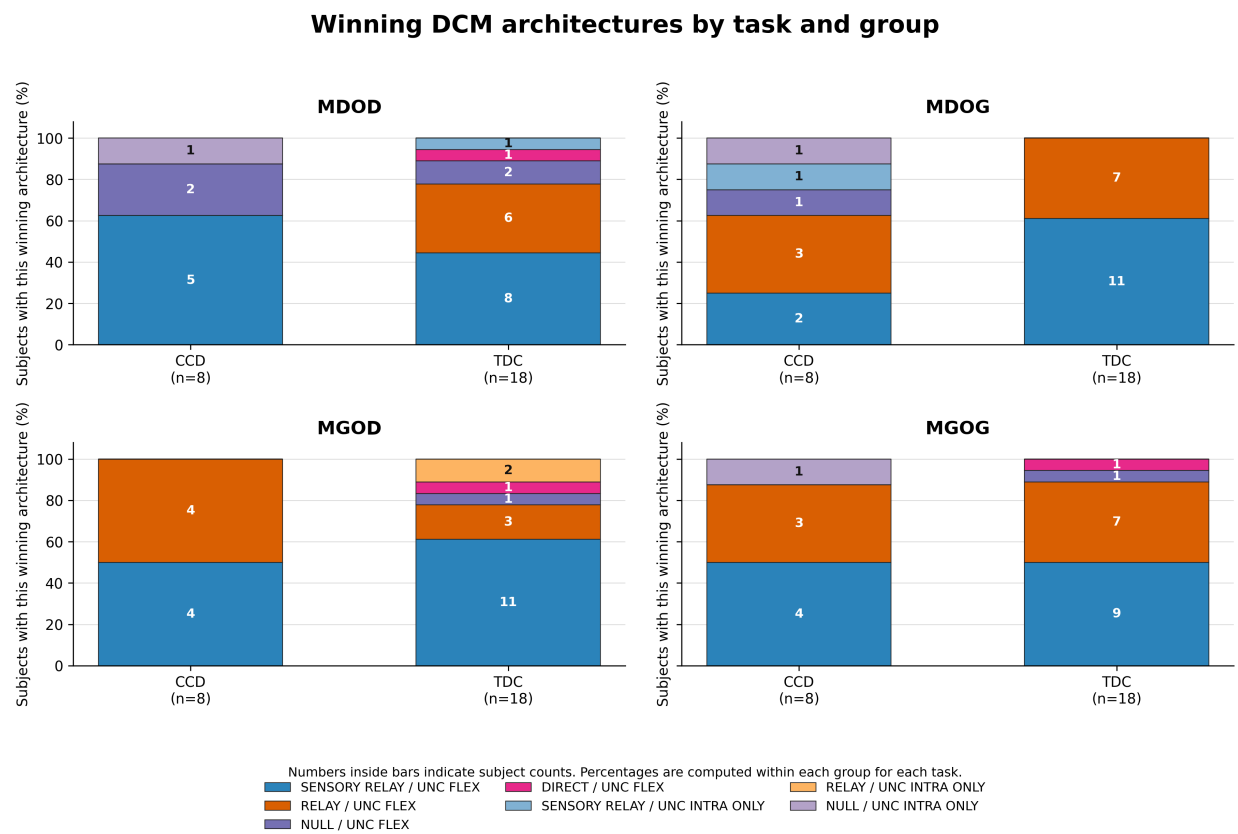


Figure 6: Distribution of winning DCM architectures across tasks and subject groups in the full cohort. Bars show the percentage of subjects within each group whose winning model corresponded to each architecture. Numbers indicate subject counts.

B Additional PEB results

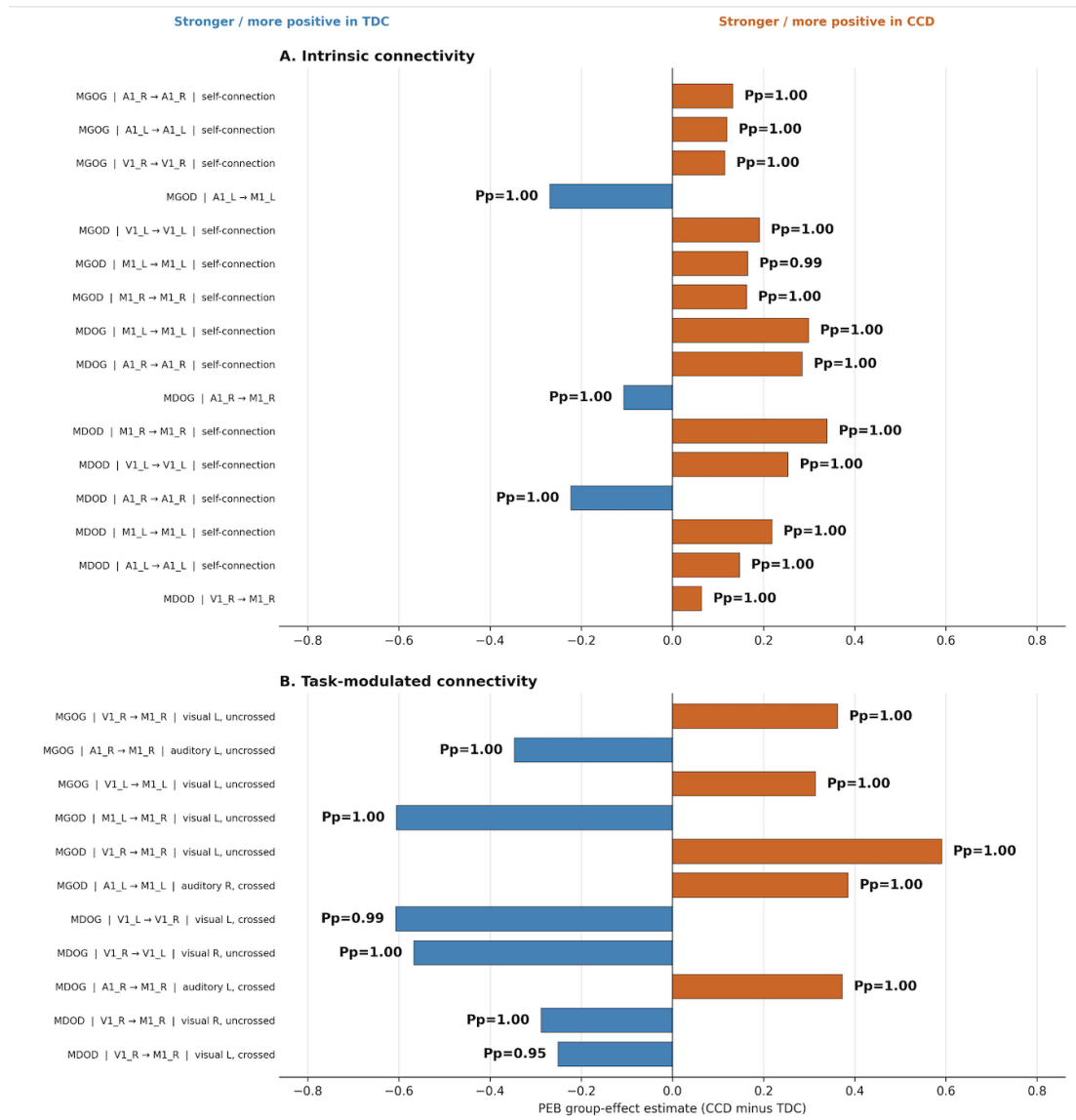


Figure 7: PEB group effects with strong posterior evidence on EC parameters in the full cohort. Positive estimates indicate parameters more positive in CCD, while negative estimates indicate parameters more positive in TDC. Only effects with $Pp \geq 0.95$ are shown.

C Switched-on DCM parameter matrices

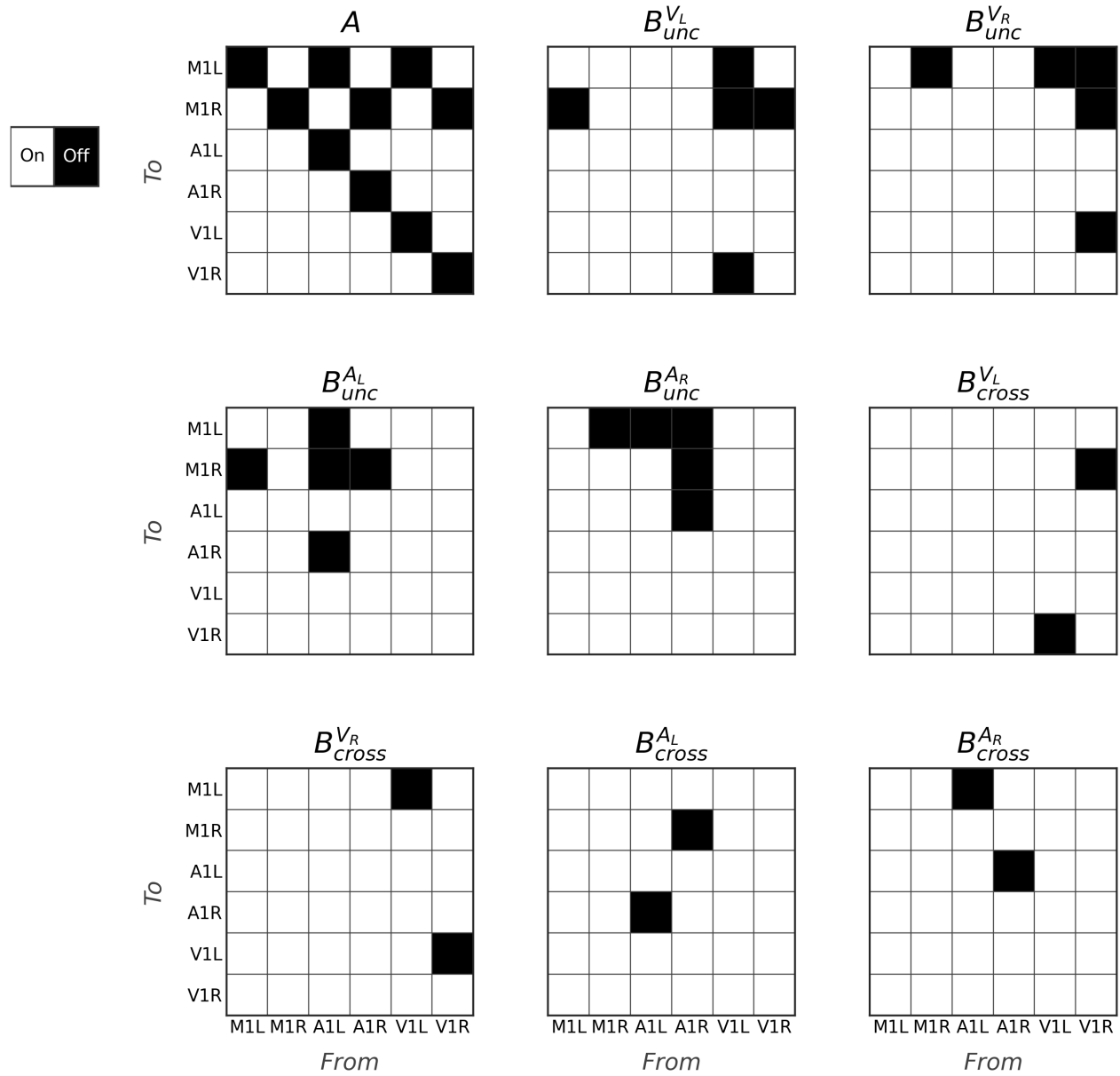


Figure 8: Switched-on DCM parameter matrices used in the PEB analysis for the dominant SENSORY_RELAY__UNC_FLEX architecture. Rows indicate target regions and columns indicate source regions. The intrinsic A matrix included 10 estimated parameters, while the condition-specific B matrices included 28 task-modulated parameters per task.

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